BNM Institute of Technology

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Internship - 2

Design Verification Using System Verilog

Verification of a 4-Request Round Robin Arbiter Using a Self-Checking SystemVerilog Test-bench

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5	System Architecture	Block diagram showing the overall design and data flow
6	Design Methodology	Explanation of the RTL design approach, modules developed, and design flow
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PROBLEM STATEMENT

- In digital systems, multiple devices often request access to a shared resource simultaneously. Without a proper arbitration mechanism, conflicts may occur, leading to unfair resource allocation and starvation of low-priority requests.
- This project aims to verify a 4-request Round Robin Arbiter using a self-checking SystemVerilog testbench. The verification environment automatically compares the expected and actual outputs, ensuring correct round robin arbitration without manual intervention.

OBJECTIVES

Design.

- Design a 4-request Round Robin Arbiter using Verilog.

Develop

- Develop a self-checking SystemVerilog verification environment.

Verify

- Verify the arbiter functionality using random test cases.

Compare

- Compare expected and actual outputs automatically using a scoreboard.

Ensure

- Ensure fair arbitration among all requesters through functional verification.

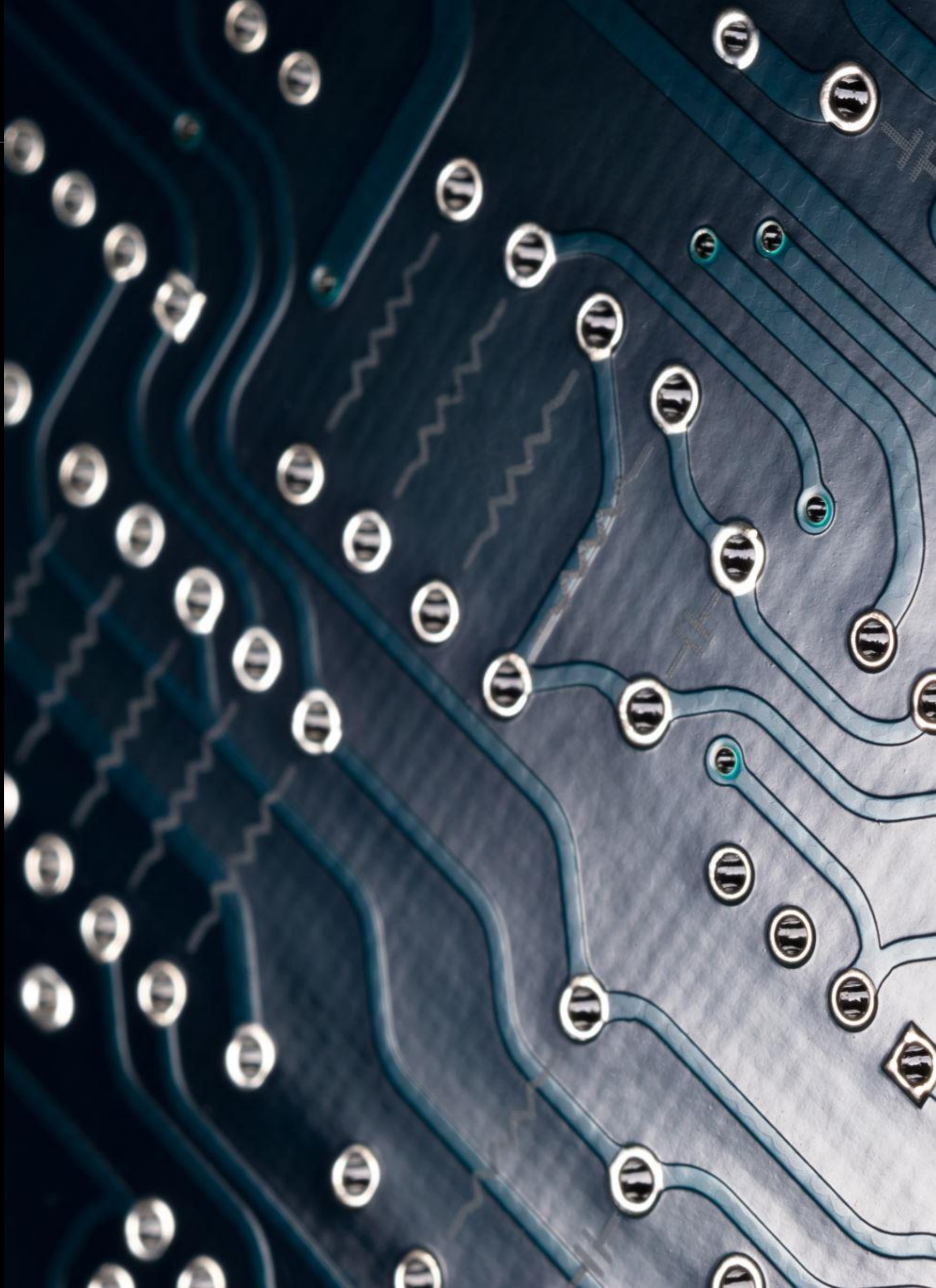
BACKGROUND

BACKGROUND

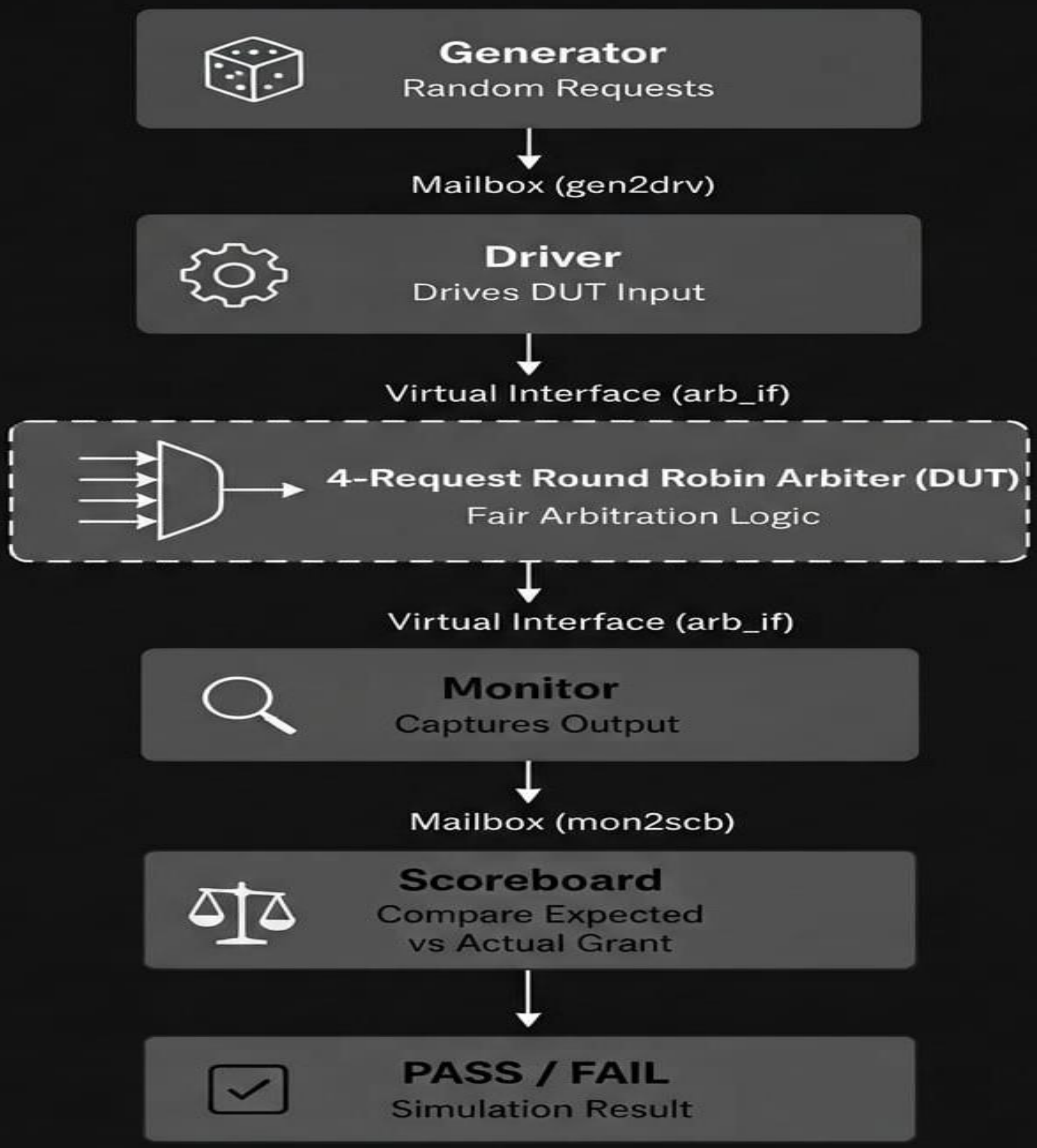
- A Round Robin Arbiter is a scheduling circuit used to provide fair access to shared resources among multiple requesters.
- Unlike fixed-priority arbiters, it changes the priority after every successful grant so that every requester gets an equal opportunity.

APPLICATIONS

- AMBA Bus Arbitration
- AXI Interconnects
- Memory Controllers
- DMA Controllers
- Network Routers
- Multi-Core Processors



SYSTEM ARCHITECTURE

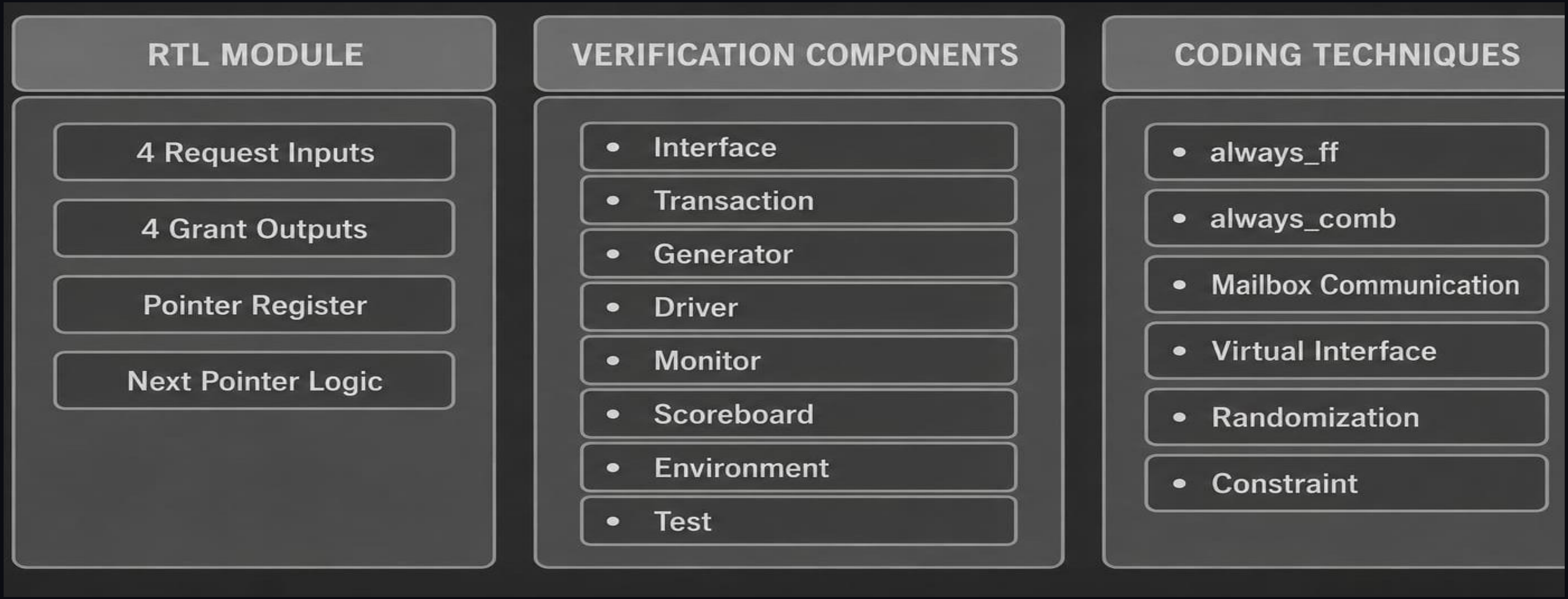


DESIGN METHODOLOGY

RTL DESIGN FLOW

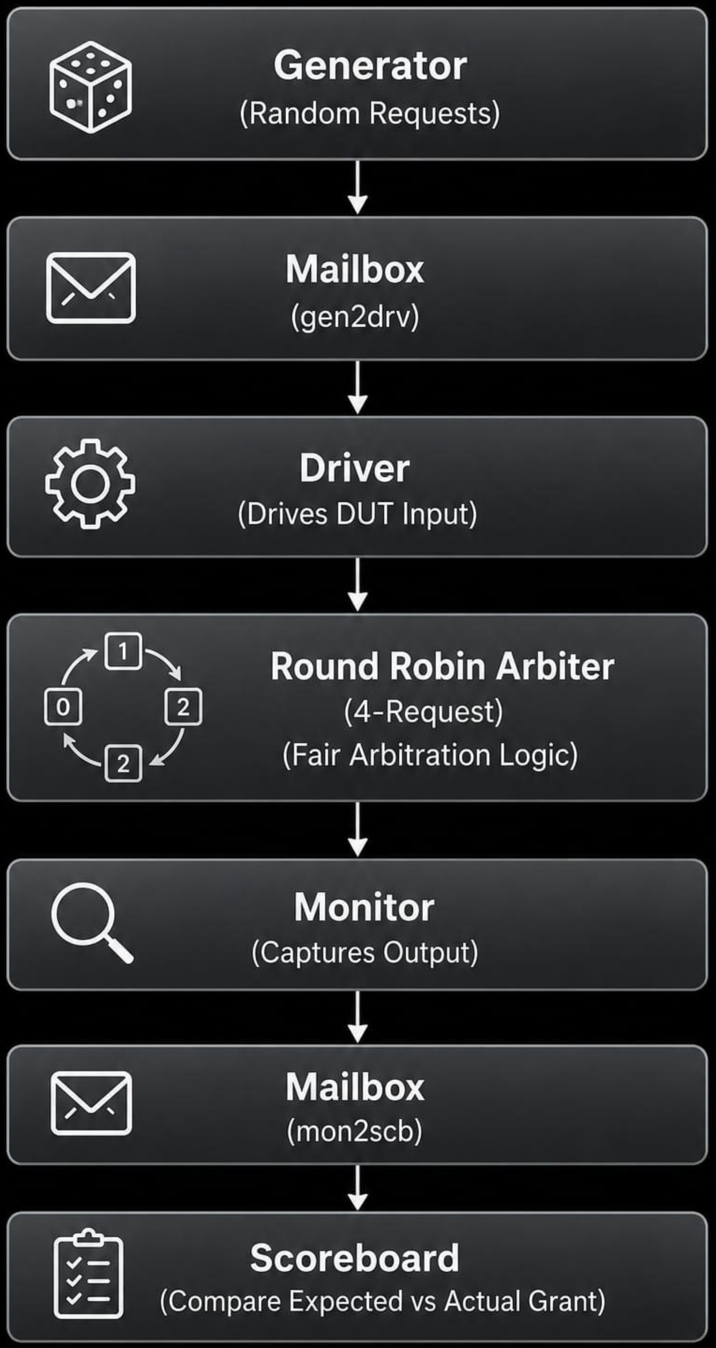
- Write RTL using Verilog.
- Simulate RTL.
- Develop SystemVerilog verification environment.
- Connect DUT with testbench.
- Apply random test vectors.
- Monitor outputs.
- Compare outputs.
- Generate PASS/FAIL.

VERILOG/SYSTEMVERILOG IMPLEMENTATION

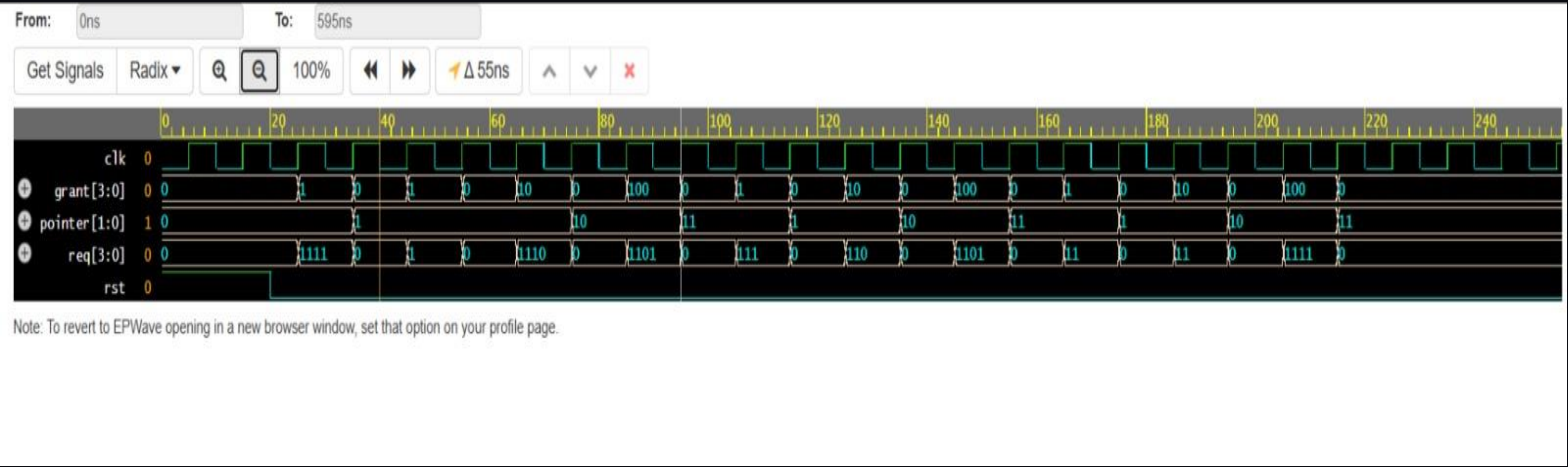


VERIFICATION METHODOLOGY

- Verification Process
- Generator creates random transactions.
- Driver applies requests.
- Monitor captures grant.
- Scoreboard predicts expected grant.
- PASS displayed if matched.
- FAIL displayed otherwise.



SIMULATION RESULTS



CHALLENGES & SOLUTIONS

CHALLENGES	SOLUTIONS
Infinite simulation	Added proper finish condition
Duplicate monitor transactions	Removed duplicate mailbox put
Scoreboard mismatch	Implemented Round Robin reference model
Randomization issue	Added constraints for valid test cases
Synchronization issue	Used mailbox and virtual interface

LEARNING OUTCOMES

- VLSI Design Flow
- Verilog RTL Coding
- System Verilog Verification
- Virtual Interfaces
- Mailboxes
- Generator, Driver, Monitor and Scoreboard Architecture
- Self-Checking Testbench
- Randomized Verification
- Debugging Simulation Errors
- Round Robin Arbitration

CONCLUSION & FUTURE SCOPE

CONCLUSION

- Designed a 4-request Round Robin Arbiter.
- Verified it using a self-checking SystemVerilog testbench.
- Validated with random test cases.
- Achieved PASS in simulation.

FUTURE SCOPE

- Add Functional Coverage.
- Implement SystemVerilog Assertions (SVA).
- Convert to UVM.
- Extend to 8/16-request and AXI4 arbiters.

THANK YOU!